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PERFORMANCE SPECIFICATION
COOLER-DEWAR GROUP, OR-332/A, OR-333/A

This performance specification is approved for use by USACECOM, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification covers only the Cooler-Dewar group to be used in the AN/ASQ-170, Acquisition Designation Sight (TADS) and the AN/AAQ-11, Pilot Night Vision Sensor (PNVS); herein referred to as the Cooler-Dewar.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other section of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Handbooks. The following handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2). Handbooks are for guidance only and therefore are not mandatory.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-454 - Standard General Requirements for Electrical Equipment.

(Unless otherwise indicated, copies of the above specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.2.2 Other Government documents, drawings and publications. The following Government documents, drawings and publications form a part of this specification to the extent specified herein.

PURCHASE DESCRIPTION

DEPARTMENT OF DEFENSE

PD-0182-001A - Cooler, Cryogenic, Long Life One Watt Integral, HD-1033D/UA

DRAWINGS

USACECOM

5008609	- Installation and Interface Dimensions, Mechanical Cryogenic cooler, HD-1033D/UA
SM-C-806033	- Bellows, Cooler
SM-D-806561	- Detector-Dewar DT-594( )/UA

(Copies of Specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.C), the samples shall be subjected to a first article inspection in accordance with 4.2.

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3.2 Materials. The contractor shall select materials that are fully capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but are not mandatory. Selection criteria for the class, grade or type part shall ensure that the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.3 Description. The Cooler-Dewar is a component of the AN/ASQ-170, Target Acquisition Designation Sight (TADS) and the AN/AAQ-11, Pilot Night Vision Sensor (PNVS), an infrared detection system used aboard military aircraft. The cooler-Dewar group consists of an infrared Detector-Dewar (DT-594/UA), a Cryogenic Cooler (HD-1033D/UA), and a thermal conductor with associated fastening hardware. This group requires further alignment prior to final installation in TADS/PNVS.

3.3.1. Common module selection. The detector and cooler components shall meet the requirements of their respective documents and be inspected in accordance with the verification assurance provisions of these documents prior to assembly of the Cooler-Dewar Group, OR-333/A (see 6.2.k).

3.3.2 Thermal conductor. The bellows thermally connects the cooler cold finger to the Detector-Dewar. The recommended parameters and installation procedure for the thermal conductor can be found on Drawing SM-C-806033. An alternate definition and installation procedure of the thermal conductor may be developed by the contractor, but shall be approved by the Government prior to use (see 6.2.b).

3.3.3 Dimensions. To ensure that items produced IAW this specification accurately interface with existing hardware, dimensions and dimensional tolerances of the Detector-Dewar and Cooler shall be in accordance with Drawings SM-D-806561 and 5008609, respectively.

3.4 Design criteria. The devices built IAW this specification shall be designed to ensure form, fit, function and interface requirements, as specified herein, are met.

3.5 Performance characteristics.

3.5.1 Defective Elements. Unless otherwise specified herein, the Detector-Dewar shall be operated at 80°K (+5, -20°K). The Detector-Dewar shall contain no more than 12 defective elements. Of the 12 defective elements, no more than 4 may be hard defects. In the central 18 elements of the array there shall be no hard defects and no more than 2 soft defects. The central 36 elements shall contain no hard defects and no more than 4 soft defects. A maximum of two hard defects may be located in elements 1-72, and a maximum of two may be located in elements 109-180. Hard defects shall be separated from one another by at least 4 elements and shall be electrically strapped to an adjacent element with common ground. Soft defects located within the central 36 elements shall be separated from any other defect by at least 4 non-defective channels.

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Soft defects shall not be strapped unless the strapping is from an adjacent hard defect at the end of a group of 5 commonly grounded elements. (Perform 4.5.1 for acceptance).

3.5.2 Detector Microphonics. When cooled to a temperature of 60-80°K, biased in accordance with paragraph 3.5.2.1 and vibrated in accordance with paragraph 3.6.3, all elements of the Detector-Dewar shall have no noise peaks above the white noise reference value greater than 6dB from 300 to 10,000 hertz (Hz) and 10 dB from 10,001 to 20,000 Hz. (Perform 4.5.2 for acceptance).

3.5.2.1 Detector bias. The bias current shall not exceed 4.0 milliamperes on any single element. The total bias power dissipated by the array shall not exceed 100 milliwatts. The detector resistor bias pack shall be operated at +5.0, ±0.2 Vdc. Each detector element shall be biased with current flow in the same direction. (Perform 4.5.2 for acceptance).

3.5.3 Cooldown time. The cooldown time to reach a detector temperature of 80°K shall be in accordance with Figure 2. (Perform 4.5.3 for acceptance).

3.5.4 Cooler input power. The total input power to the cooler shall not exceed that shown in Figure 3 after the detector temperature has reached at least 80°K. The input voltage shall be 117 ± 2 volts alternating current (VAC), 400 ± 20 hertz (Hz) power source. Perform 4.5.4 for acceptance.

3.6 Environmental conditions.

3.6.1 Temperature extremes. The Cooler-Dewar Group shall not be damaged when operated at temperatures from -32°C to +60°C and stored at temperatures from -54°C to +71°C.

3.6.2 Temperature shock. The Detector-Dewar Group shall not be damaged by sudden changes in temperature between -54°C and +71°C .

3.6.3 Vibration exposure. The Cooler-Dewar Group shall not be damaged when exposed to the vibration levels and amplitudes shown in Figure 1 for a duration of 15 minutes in each of the three orthogonal axes. The requirement of 3.5.2 shall also be maintained while subjected to vibration.

3.7 Workmanship. All electronic parts, components, assemblies or subassemblies shall be free of smudges, excess solder, metal chips or the existence of any foreign material on any surface. Bearing assemblies shall be free from rust, dirt, or tool marks. Wires and integrated circuitry shall be protected from contact with rough or irregular surfaces and be shielded from shorting.

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4. VERIFICATION

4.1 Classification of inspections.

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor. Reliability verification is required only on the Cooler during first article inspection (see 3.1 and 6.2.c.).

4.2.1 Visual examination. All materials, parts, processes and assemblies shall be visually examined in accordance with Table I for conformance to the applicable specification or drawing.

TABLE I. Inspections.

Inspection	Requirement paragraph	Verification paragraph
Any part or component missing or damaged	3.2	4.2.1
Workmanship not as specified	3.7	4.2.1

4.2.2. Tests. Upon successful completion of the visual examination specified in 4.2.1, four samples shall be subjected to all inspections listed in Table II. First article Coolers must successfully meet the requirements of the respective Cooler specification. Inspections may be conducted in any order. Failure of any test shall constitute failure of the first article (See 6.2). Sampling shall be in accordance with ANSI/ASQC Z1.4.

4.2.3 Disposition of first article samples. Sample units, which have been subjected to and passed first article inspections, may be accepted on the contract or purchase order provided sample units are resubjected to and pass Group A inspections. The sample reliability first article Coolers shall not be considered as part of the procurement quantities.

TABLE II. First article inspection.

Inspection	Requirement paragraph	Verification Paragraph
Defective elements	3.5.1	4.5.1
Detector microphonics	3.5.2	4.5.2
Cooldown time	3.5.3	4.5.3
Cooler input power	3.5.4	4.5.4
Environmental conditions	3.6	4.6

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4.3 Conformance inspection. The conformance inspection shall include the examination of 4.3.1 and the Group inspections of 4.4.

4.3.1 Examination. Each Cooler-Dewar assembly shall be examined for compliance with the requirements specified in 3.5.1 through 3.7. Any redesign or modification of the contractor's standard product to comply with specified requirements or any necessary redesign or modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Group inspections.

4.4.1 Group A inspections. Group A inspection shall be conducted on all Cooler-Dewar groups. Group A tests listed in Table III may be performed in any order. Failure of any test shall be cause for rejection of that unit.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Verification paragraph
Defective elements	3.5.1	4.5.1
Detector microphonics	3.5.2	4.5.2
Cooldown time	3.5.3	4.5.3
Cooler input power	3.5.4	4.5.4

4.4.2. Group B inspection. Not required.

4.4.3 Group C inspection. Group C inspection shall be conducted on Cooler-Dewar groups selected from units which have passed the tests in 4.4.1. The environmental tests of 4.6 shall be verified on a sample quantity defined in the contract or purchase order. Unless otherwise specified in the contract or purchase order, samples shall be randomly selected from units produced during sampling period. Sampling schedule ias follows:

MONTHLY LOT SIZE	NUMBER OF UNITS TO BE TESTED
1-9	1
10-29	2
30 or more	3

4.4.3.1 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order (see 6.2.e).

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4.4.3.2 Re-inspection of conforming Group C samples. Unless otherwise specified, the sample units which have been subjected to and passed Group C inspection may be accepted on the contract or purchase order provided that all visible and known damage is repaired and sample units are re-subjected to and pass Group A inspections.

4.4.4 Group D inspection. Not required.

4.5 Methods of inspection. The contractor/supplier may use his own test methods, measuring devices, tools, fixtures and other equipment during the conduct of verification testing. All of the performance requirements per Section 3 shall be fully verified as being met.

4.5.1 Defective elements. To measure compliance with 3.5.1, the detector array shall be checked for continuity with an ohm meter or similar device with current limited to one (1) ma. Lack of continuity indicates an open element which will be cause for rejection. Failure to meet the requirements of 3.5.1 shall constitute failure of this test.

4.5.2 Detector microphonics. Verify compliance with the performance characteristics of 3.5.2 and 3.5.2.1 with a noise power spectrum analyzer or similar device. Failure to meet the requirements of 3.5.2 and 3.5.2.1 shall constitute failure of this test.

4.5.3 Cooldown time. Verify compliance with the performance characteristic of 3.5.3 by measuring the cooldown time in accordance with Figure 2. Failure to meet the requirement of 3.5.1 shall constitute failure of this test.

4.5.4 Cooler input power. Verify compliance with 3.5.4 by using an ohm meter or similar device and comparing these values with the requirements indicated in Figure 3. Failure to meet the requirements of 3.5.4 shall constitute failure of this test.

4.6 Environmental tests. Environmental testing methods contained in 6.2. d are acceptable methods for determining compliance with Section 3.6 et al. The contractor may elect to use alternative methods and testing procedures. In such cases, the contractor shall be required to show where his own tests and measurements for verifying the environmental conditions are equivalent to the methods cited herein. When the contractor desires to deviate from these cited methods, he shall be prepared to show equivalency via a test plan which has been approved by the contracting office. Also, environmental tests shall be accomplished using known testing apparatus such as an environmental test chamber, vibration test table, pendulum shock device, or similar test equipment for which the technical capabilities to properly simulate extreme climatic or transportation conditions are scientifically documented. The methods for testing to verify compliance with all stated performance characteristics selected by the contractor shall be equivalent to the recommended procedures found herein.

4.6.1 Temperature extremes. Verify that the Cooler-Dewar meets the requirements of paragraph 3.6.1. A recommended method for verification may be found in 6.2. Failure to meet the requirements of 3.6.1 shall constitute failure of this test.

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4.6.2 Temperature shock. Verify that the Cooler-Dewar meets the requirements of paragraph 3.6.2. A recommended method for verification may be found in 6.2. Failure to meet the requirements of 3.6.2 shall constitute failure of this test.

4.6.3 Vibration. Verify that the Cooler-Dewar meets the requirements of paragraph 3.6.3. A recommended method for verification may be found in 6.2. Failure to meet the requirements of 3.6.3 shall constitute failure of this test.

5. PACKAGING

5.1 Packaging and Marking. For acquisition purposes, the packaging and marking requirements shall be as specified in the contract or purchase order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The Cooler-Dewar Group is intended for use in the TADS and PNVS.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. The thermal conductor assembly procedure should be submitted to the Contracting Officer of the Acquisition Activity for approval.

c. First article tests are required for inspection and approval, and the number of units required (see 3.1).

(1) Time frame for submission of first article test reports.

(2) Time frame for approval of first article test reports.

d. Alternate use of commercial environmental test methods in lieu of MIL-STD-810 for

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environmental testing as approved by the contracting officer. The following methods of MIL-STD-810 apply:

(1) 501.1 and 502.1, Procedure I, for High and Low temperatures, respectively (3.6.1).

(2) 503.1, Procedure I for Temperature shock (3.6.2).

(3) 514.2, Procedure IV for Vibration (3.6.3).

e. Production delivery schedule.

(1) Defined in terms of monthly lots.

(2) Actions required relative to Group C failures.

(3) Disposition/refurbishment of Group C samples.

f. The contractor will order Group A detector inspection reports for use in validation of the requirements of section 3, et al. The Contractor will also validate conformance with Group A, C and D inspections for both the Detector-Dewar, DT-594/UA and the Cooler, Cryogenic, Long Life per paragraph 3.3.1, Common Module selection.

g. When first article rough handling tests are not required.

h. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packing.

i. Specify drawing 54490-5008607-2 for HD-1033D/UA-Cooler-Dewar subassembly, 180 element.

j. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.2.2).

k. Provisions for inspecting DT-594/UA Detector-Dewar and HD-1033D/UA Cryogenic Cooler, Long Life, one watt integral.

6.3 Definitions.

6.3.1 Damage. Any breakage, loosening, shifting, evidence of corrosion, failure of any finish, hardware, connection or components, or any deterioration in quality, or failure to operate within the specified requirements.

6.3.2 Element. One (1) channel in the 180 element detector array.

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6.3.3 Inspection lot size. An inspection lot is defined as three months continuous production or 120 units, whichever comes first, of those assembles which have successfully passed Group A inspection.

6.3.4 Strapping. Strapping is defined as electrically connecting a defective element to a nondefective element. The pair of strapped elements thus formed, exhibiting 100 percent crosstalk, shall be considered as a single defect.

6.3.5 Defective element. A defective element will be defined as an element which fails to meet any one of the performance requirements of the Detector-Dewar DT-594/UA, specifically paragraphs 3.6.5, 3.6.5.1, 3.6.6, 3.6.7, and 3.6.14 of MIL-PRF-49172 at the time of delivery from the Detector-Dewar vendor.

6.4 Provision for inspecting.

Detector-Dewar	DT-594/UA	MIL-D-49172
Cryogenic Cooler, Long Life, One watt Integral	HD-1033D/UA	PD-0182-001A(CR)

6.5 Subject term (key word) listing.

Cryogenic
Night Vision
Infrared

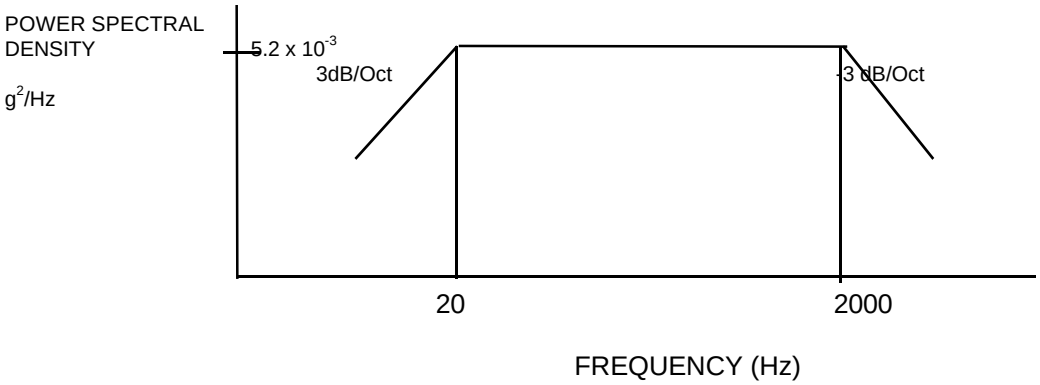
6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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NOTES:

1. Curve limiting PSD response to +3 dB.
2. The RMS value is 3.2g.

FIGURE 1. Random Vibration.

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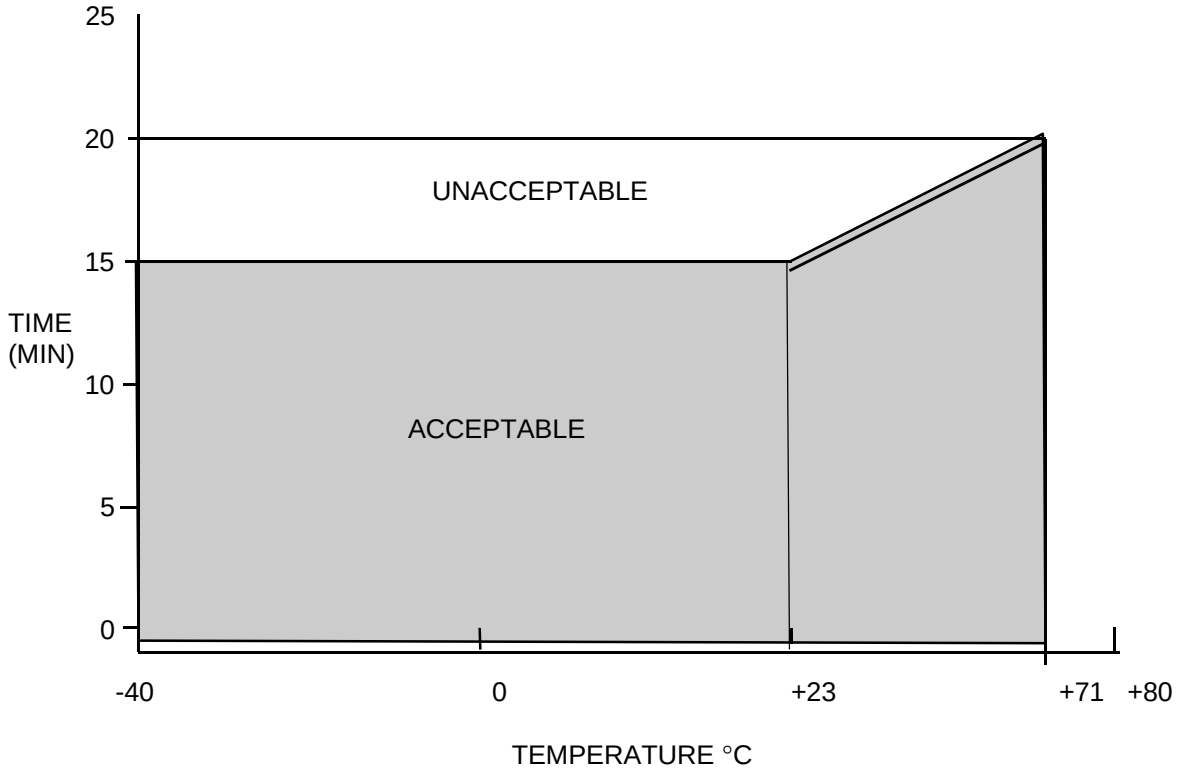


FIGURE 2. Cool Down Curve.

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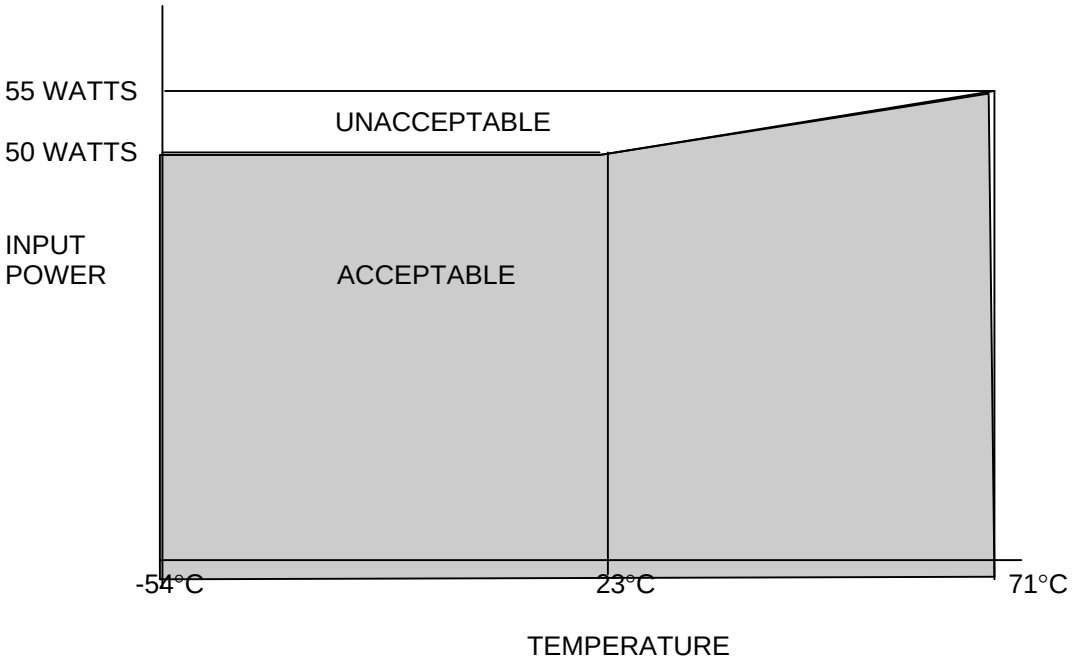


FIGURE 3. Input Power With Heat Load Applied.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF49307A	**2. DOCUMENT DATE (YYMMDD)** 3 July 1997

3. DOCUMENT TITLE COOLER-DEWAR GROUP, OR-332/A, OR-333/A

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732) 532-9913 (2) AUTOVON 992-9913
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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